SPECTER LABS PREPRINT

Where Tissues Break: Compositional Prediction of Morphogenetic Vulnerability

Wiring diagram factorization reveals parameter-dependent transitions in tissue fragility

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April 2026

ABSTRACT

When a 1D reaction-diffusion tissue is severed, which cut causes the most damage? The balanced-bipartition heuristic says the midpoint. We show this is wrong at every parameter value tested: the most damaging cut is determined by *fragment viability* — whether each resulting piece exceeds the minimum domain size for pattern maintenance. Using a compositional formalization of the Grodstein–McMillen–Levin (2023) closed-loop morphogenesis model, we factorize severed tissues into independent fragments via wiring diagram decomposition and rank cuts by a normalized severity functional combining peak-count shift, concentration-profile distance, and shape-string edit distance. In a 30-cell chain scanned over 100 values of the activator diffusion coefficient D_a, the identity of the worst cut changes abruptly at two transition points, and the margin between the top two cuts collapses near these transitions. An RD-only ablation (no wave, no controller) confirms the effect is driven by fragment viability in the reaction-diffusion layer alone, not by the directional computation wave.

Keywords. morphogenesis, polynomial functors, wiring diagrams, reaction-diffusion, bioelectricity, tissue vulnerability

1. Introduction

A developing embryo is remarkably robust. Scramble the organs of a frog tadpole's face and the cells rearrange themselves back to the correct configuration [1]. Cut a planarian flatworm into pieces and each fragment regenerates a complete animal. Disrupt gap junctions — the electrical synapses connecting cells — and the same wild-type genome can produce one-headed or two-headed worms, with the outcome stored as a stable bioelectric pattern persisting across unlimited generations of re-cutting [2, 3].

These results, primarily from the laboratory of Michael Levin, established that tissues store morphogenetic goals in bioelectric patterns, not just in DNA [4]. The patterns are rewritable: a brief voltage signal can induce eye formation in the gut [5], and a 24-hour bioreactor stimulus triggers 18 months of limb regeneration [6]. Formalizing these competencies requires computable morphospaces where the topology, metric, and barrier structure are known—where "navigation" has a precise meaning and attractor discreteness can be studied as a topological feature. Crucially, mapping the barriers and boundaries of this space requires understanding where the system fails.

A natural question follows: if tissues are robust to many perturbations, which perturbations push them outside their viable morphospace? If you sever a tissue at a particular location, how bad is the damage, and does the answer depend on the tissue's biochemical state?

We show that in a specific 1D closed-loop morphogenesis model, the most damaging cut position is determined not by connectivity but by *fragment viability* — whether each resulting piece is large enough to sustain its reaction-diffusion pattern. The critical fragment size depends on the diffusion parameters, so the vulnerability landscape changes with the tissue's biochemical state. We find transition points where the identity of the most vulnerable location shifts abruptly, and parameter regimes where the tissue is uniformly fragile (many cuts are nearly equally damaging).

We arrive at this result by formalizing the closed-loop reaction-diffusion system of Grodstein, McMillen & Levin (2023) [7] using polynomial functors and wiring diagrams from applied category theory. The formalization makes the key operation — factorizing a severed tissue into independent fragments — a natural consequence of the compositional structure.

Section 2 reviews the biological and mathematical context. Section 3 defines the severity functional, the baseline heuristic, and the factorization. Section 4 presents the vulnerability landscape and ablation experiments. Section 5 interprets the results and identifies testable predictions.

2. Background

2.1. Bioelectric morphogenesis

Cells communicate through bioelectric signals: voltage gradients established by ion channels and propagated through gap junctions [8]. In many organisms, these patterns encode target morphology [9].

Key experimental findings relevant to this work:

- **Planarian head-number bistability.** The same wild-type genome supports at least two stable morphologies (one head or two heads), selected by the bioelectric state [2].

- **Gap junction disruption induces morphological change.** Blocking gap junctions switches planaria between morphological states [10]. The *wiring*, not just the cells, determines the outcome.
- **Cancer as connectivity failure.** Disrupting bioelectric connectivity causes cells to revert to unicellular-scale behavior [11, 12]. Restoring connectivity suppresses oncogene-driven tumors in Xenopus embryos [13].

All published experimental work uses *global* interventions (bath application of pharmacological agents, whole-organism genetic knockdown). No study has systematically varied *where* in the tissue the disruption occurs. Our computational result addresses this gap.

2.2. Reaction-diffusion pattern formation

Turing (1952) showed that two chemicals — an activator promoting its own production and an inhibitor suppressing it — can form spatial patterns if the inhibitor diffuses faster [14]. The patterns have characteristic wavelength $\lambda_{\mathrm{RD}} \approx \sqrt{D_A/K_{D,A}}$, and a domain of length L supports approximately L/λ_{RD} peaks [15].

There exists a *minimum domain size* below which no pattern can form [16]. A fragment shorter than approximately one wavelength converges to a uniform state or a qualitatively different pattern. This minimum-domain-size effect is the mechanism behind our main result.

2.3. The Grodstein model

Grodstein, McMillen & Levin (2023) constructed a closed-loop morphogenesis model with three components [7]:

1. **Reaction-diffusion (RD) layer.** A 1D chain of n cells with activator (A_i) and inhibitor (I_i) concentrations, coupled by nearest-neighbor diffusion.
2. **Computation wave.** A gene regulatory network (GRN) in each cell uses Schmidt-trigger logic to count peaks as a wavefront propagates from tail (cell 1) to head (cell n).
3. **Feedback controller.** Reads the peak count, compares to target N, adjusts the diffusion scale D_a, and repeats until the count matches.

The RD dynamics for each cell are:

$$\frac{dA_i}{dt} = g_A \cdot f(A_i, I_i) - d_A \cdot A_i + D_a \cdot \Delta_h A_i$$

$$\frac{dI_i}{dt} = g_I \cdot f(A_i, I_i) - d_I \cdot I_i + D_i \cdot \Delta_h I_i$$

where $f(A, I) = 1/(1 + (I/(A + \varepsilon))^{n}_{H})$ is a Hill-ratio activation function and $\Delta_h A_i = (A_{i-1} + A_{i+1} - 2A_i)/h^2$ is the discrete Laplacian with spacing $h = L/n$. Neumann boundary conditions: $A_0 = A_1$, $A_{n+1} = A_n$. The head-tail asymmetry is imposed by a preseed: cells near cell n (head) are initialized to high activation, cells near cell 1 (tail) to zero.

2.4. Polynomial functors and wiring diagrams

A polynomial functor [17] has the form $p(y) = \sum_{s \in S} y^{R(s)}$, where S is a set of states and $R(s)$ is a set of responses available at state s. A cell in the Grodstein model fits this structure: in RD mode it exposes diffusion ports, in wave mode it exposes signal ports, in done mode it exposes a peak count.

A *wiring diagram* [18] specifies how polynomial functors compose. For this paper, the critical operation is: severing the tissue at position k removes the edges between cell k and cell $k+1$ in the wiring diagram. The resulting diagram decomposes into two independent subdiagrams, and the dynamics factorize.

Figure 1: Cell numbering convention. Cell 1 (tail, low initial activation) through cell n (head, high initial activation). A cut at position k severs the edge between cells k and $k+1$, producing a left fragment of k cells and a right fragment of $n-k$ cells. *[To be replaced with actual schematic.]*

3. Methods

3.1. Tissue factorization

> **Proposition (Factorization).** Let W be the wiring diagram of a 1D chain of n cells, and let W_k denote the diagram with the edge between cells k and $k+1$ removed. Then W_k decomposes as $W_k = W_L \otimes W_R$ where W_L is the subdiagram on cells $\{1, ..., k\}$ and W_R on cells $\{k+1, ..., n\}$. The composed dynamical system factorizes: each fragment settles to its own attractor independently.
>
> *Proof.* In the 1D chain, the only coupling between cells k and $k+1$ is through the diffusion terms $D_a(A_{k+1} - A_k)/h^2$ and $D_i(I_{k+1} - I_k)/h^2$. Removing this edge sets the Neumann condition $A_{k+1} = A_k$ at the boundary of each fragment. The resulting ODEs for cells $\{1, ..., k\}$ depend only on states within that set, and likewise for $\{k+1, ..., n\}$. $\square$

This factorization is exact for the 1D topology. We validate it computationally by comparing factorized predictions against direct simulation of the full severed chain for every candidate cut. In all cases tested, the results agree.

The factorization extends to multiple cuts: cutting at positions $k_1 < k_2$ produces three independent fragments, and the dynamics factorize into three independent systems.

3.2. Severity functional

For each candidate cut, we measure the phenotype shift between the uncut tissue's settled pattern and the concatenated pattern of the independently settled fragments. The severity score combines three normalized components with equal weight:

$$S(k) = \frac{1}{3}\left(\hat{\Delta}_{\text{peak}(k)} + \hat{\Delta}_{\text{profile}(k)} + \hat{\Delta}_{\text{shape}(k)}\right)$$

where:

- $\hat{\Delta}_{\text{peak}(k)} = |p_k - p_0| \ / \max_j |p_j - p_0|$ is the normalized absolute peak-count change (p_0 = uncut peak count, p_k = total peaks in fragments after cut at k)
- $\hat{\Delta}_{\text{profile}(k)} = \overline{|A^k - A^0|} / \max_j \overline{|A^j - A^0|}$ is the normalized mean L1 distance between concentration profiles
- $\hat{\Delta}_{\text{shape}(k)} = d_{\text{Lev}(\sigma_k, \sigma_0)} / \max_j d_{\text{Lev}(\sigma_j, \sigma_0)}$ is the normalized Levenshtein distance between shape strings

Shape strings are symbolic representations of the concentration profile: "L" marks a local minimum region, "H" marks a local maximum (see Grodstein et al. [7] for details). Each component is normalized to $[0, 1]$ by dividing by the maximum across all candidate cuts, so $S(k) \in [0, 1]$ with $S = 1$ for the worst cut.

Ties are broken by connectivity loss, then by lexicographic cut label.

3.3. Baseline: balanced-bipartition heuristic

The baseline heuristic ranks cuts by the number of *disconnected cell pairs* they create:

$$d(k) = k(n - k)$$

which is maximized at $k = \lfloor n/2 \rfloor$ (the midpoint). This corresponds to the graph-theoretic intuition that the most damaging cut maximizes the disruption to global signaling. For a single cut in a 1D chain, this is the only sensible connectivity-based ranking, since every cut removes exactly one edge.

3.4. Implementation

Each cell is a polynomial functor implemented as a `PolynomialObject` whose `readout` and `update` functions derive the active interface from the cell's state. Tissues are compiled into Catlab [19] wiring diagrams. The execution engine routes signals through the diagram's wire structure.

The RD layer uses the Tsit5 integrator from OrdinaryDiffEq.jl with absolute and relative tolerances of 10^{-8}. We verified that the categorical execution path produces identical dynamics to a hand-written coupled ODE system at machine precision.

4. Results

4.1. The worst cut is not the midpoint

In a 30-cell chain at the reference parameter $D_a \approx 9.85$, the balanced-bipartition heuristic ranks cut 15 (midpoint) first with $d(15) = 15 \times 15 = 225$ disconnected pairs. The severity ranking places cut 24 first, with $S(24) = 1.0$.

Cut 24 isolates a 6-cell *head* fragment (cells 25–30). This fragment is below the critical domain size for sustaining the original pattern and converges to a qualitatively different attractor. The 24-cell body (cells 1–24) retains the original pattern with minor boundary adjustment. The midpoint cut produces two 15-cell fragments both above the viability threshold.

Figure 2: Severity $S(k)$ vs cut position k at three representative D_a values: $D_a = 7.5$ (low, solid), $D_a = 8.5$ (mid, dashed), $D_a = 10.0$ (high, dotted). The peak of each curve is the worst cut. The midpoint ($k = 15$, vertical gray line) is never the maximum. *[Placeholder for actual plot.]*

4.2. Transition points in the vulnerability landscape

We scanned 100 values of D_a from 7.0 to 12.5 on a 30-cell chain. The identity of the worst cut changes at two transition points:

D_a range	Worst cut k^*	Fragment isolated	Top-2 margin
7.0 – 8.0	$k^* = 4$	6-cell head (cells 25–30)	0.12
8.0 – 8.25	*transition*		
8.25 – 9.25	$k^* = 25$	5-cell head (cells 26–30)	0.006 – 0.0004
9.25 – 9.5	*transition*		
9.5 – 12.5	$k^* = 24$	6-cell head (cells 25–30)	0.006

Table 1: Transition points in the 30-cell vulnerability landscape. The worst cut shifts from position 4 (isolating a short tail fragment) to positions 25/24 (isolating short head fragments) as D_a increases. The top-2 margin (severity difference between the worst and second-worst cut) collapses near the transitions. The balanced-bipartition heuristic ranks cut 15 first at every D_a value.

Figure 3: Heatmap of severity $S(k)$ over (D_a, k) space. Horizontal axis: D_a from 7.0 to 12.5 (100 values). Vertical axis: cut position k from 1 to 29. Color: severity score. The transitions appear as discontinuities in the ridge line (locus of k^*). *[Placeholder for actual heatmap.]*

Figure 4: Top-2 margin vs D_a. The margin collapses near $D_a \approx 8.1$ and $D_a \approx 9.4$, indicating parameter regimes where the tissue is uniformly fragile: no single cut is much worse than the rest. *[Placeholder for actual plot.]*

The three regimes have a clear mechanistic interpretation:

- **Low D_a (tight patterns, many peaks)**: the characteristic wavelength is short. A 4-cell tail fragment (cells 1–4) is below the viability threshold. Large margin: this cut is clearly worse than all others.
- **Intermediate D_a**: the wavelength is longer. Multiple cut positions produce fragments near the viability boundary. The margin collapses — the tissue is uniformly fragile.
- **High D_a (broad patterns, few peaks)**: a short head fragment (5–6 cells) is below threshold. The margin reopens.

4.3. Ablation: RD-only

To test whether the effect is driven by fragment viability in the RD layer or by the directional computation wave, we repeated the scan with the wave and controller removed: settle the RD at each fixed D_a, sever, settle each fragment, and compute severity.

Figure 5: RD-only ablation. Severity $S(k)$ vs cut position at $D_a = 7.5$ (left) and $D_a = 10.0$ (right), with wave and controller removed. The asymmetry persists: the worst cut is near the ends, not the midpoint. *[Placeholder for actual plot. This ablation must be run before the paper can make the mechanistic claim.]*

[Note: This figure requires running the RD-only ablation experiment, which has not yet been performed. The result will either confirm that fragment viability alone produces the asymmetry (strengthening the mechanistic claim) or reveal that the directional wave contributes (requiring the discussion to be qualified). Either outcome is informative.]

4.4. Multi-cut families

For double cuts (two simultaneous severances creating three fragments), the severity ranking diverges from the balanced-bipartition heuristic more sharply. In a validated 20-cell probe:

- Balanced bipartition ranks cuts [7, 14] first ($d = 7 \times 7 + 7 \times 6 = 91$ disconnected pairs from three fragments)
- Severity ranking places [5, 15] first: two 5-cell fragments, both near the viability boundary

Both cuts produce the same peak count, but the severity functional captures that two marginal fragments cause more total phenotype shift than one marginal and one comfortable fragment.

4.5. Scaling

In a validated 100-cell subset (cuts at positions {50, 75, 80, 85, 90}):

- Balanced bipartition ranks cut 50 first ($d = 50 \times 50 = 2500$)
- Severity ranking places cut 80 first (20-cell head fragment near viability boundary)

This is a subset probe, not a full sweep. A complete 100-cell scan is computationally feasible but was not completed for this draft.

5. Discussion

5.1. Fragment viability as the mechanism

The central finding is that morphogenetic vulnerability in this model is determined by fragment viability, not connectivity. The most damaging cut creates a fragment below the critical domain size for pattern maintenance. This critical size depends on the RD parameters (D_a, D_i, reaction rates), so the vulnerability landscape is parameter-dependent.

This mechanism explains the three properties of the vulnerability landscape: the worst cut is asymmetric because the critical fragment is at the chain boundary where the preseed creates an asymmetry; the worst cut changes abruptly because the critical domain size is a threshold; the margin narrows at transitions because multiple cut positions produce fragments near the threshold.

5.2. Relationship to existing work

The ingredients for this result exist in separate literatures: minimum domain size for Turing patterns [16], mode selection on bounded domains [20], gap junction disruption [2, 10], and critical fragment size for regeneration [21]. What has not been done is combining these into a vulnerability ranking over (parameter, cut-position) space. The compositional framework makes this combination natural.

The closest related work is Bassett & Van Gorder (2022) on landscape fragmentation and Turing patterns [22], which maps boundaries in (patch size, diffusion) space for pattern existence in fragmented ecological landscapes. Their question (does a pattern form?) differs from ours (which cut is worst?), but the mathematical structure — boundaries from domain-size thresholds — is analogous.

5.3. Predictions for experimental biology

If this result extends beyond this specific model — a hypothesis, not a claim — it suggests:

Position-dependent vulnerability. Gap junction disruption at specific locations should cause more morphogenetic damage than at other locations, even when the same number of connections is broken.

Parameter-dependent vulnerability maps. As bioelectric parameters change, the vulnerability map should reorganize. The reorganization should be abrupt, not gradual.

These predictions could be tested by spatially targeted gap junction disruption (optogenetic manipulation of connexin/innexin expression) combined with voltage imaging.

5.4. The role of the categorical formalization

The factorization of a severed 1D chain into independent fragments is obvious in ODE language: removing the coupling terms between cells k and $k + 1$ makes the two subsystems independent. The wiring diagram formalization adds structure in three ways:

1. It makes the factorization a *typed* operation: the wiring diagram decomposes into subdiagrams with validated port compatibility, not just an ad hoc splitting of a state vector.
2. It generalizes to non-chain topologies where the factorization is less obvious (2D grids with complex boundary geometry).
3. It enables the severity ranking to be computed from the *composition structure* rather than from simulation: each fragment's attractor can be predicted from the subdiagram and the RD parameters.

Whether this structural advantage justifies the categorical machinery for 1D chains is debatable. The payoff is expected to increase in higher dimensions and with richer wiring topologies.

5.5. Limitations

1. **1D chain topology.** Real tissues are 2D/3D. The exact factorization holds for 1D but is approximate for higher dimensions.
2. **Specific RD model.** We use a Werner Hill-ratio model. The qualitative result should hold for any Turing-type system with a critical domain size, but specific transitions depend on the model.
3. **No experimental validation.** All results are computational.
4. **Staged execution.** The three phases execute sequentially. Real cells may operate concurrently.
5. **Sensitivity to the severity functional.** The ranking may depend on the choice of metric and weights. A sensitivity analysis across alternative functionals (peak-count only, L2 distance, shape distance only) is needed and has not yet been performed.
6. **Ablation incomplete.** The RD-only ablation described in Section 4 has not yet been run. Until it is, the mechanistic claim (fragment viability alone, not directional wave) is a hypothesis.

6. Reproducibility

Code and documentation for this addendum are in the public repository `https://github.com/Specter-Research-Labs/research-registry`, under `addenda/poly-morphogenesis/`.

From a fresh clone:

```
it
```

Key demo commands:

```
it
```

The implementation provides:

- Source code in `src/` (RD dynamics, wave counting, controller, Poly layer, compilation, wiring diagrams)
- Comprehensive test suite in `test/` covering all major components
- Documentation and diagrams in `docs/`
- Source-aligned implementations of RD, wave counting, and controller phases matching the Grodstein et al. (2023) paper

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